Assignment2

2026-10-04

Part 1

Use the 'divorce_margarine' dataset from the 'dslabs' package to explore the relationship between margarine consumption and divorce rates in Maine. Visualise the data, make a correlation test. Report the correlation coefficient and p-value in APA format, and briefly discuss the results in a practical way

```
# Downloading data
data("divorce_margarine")

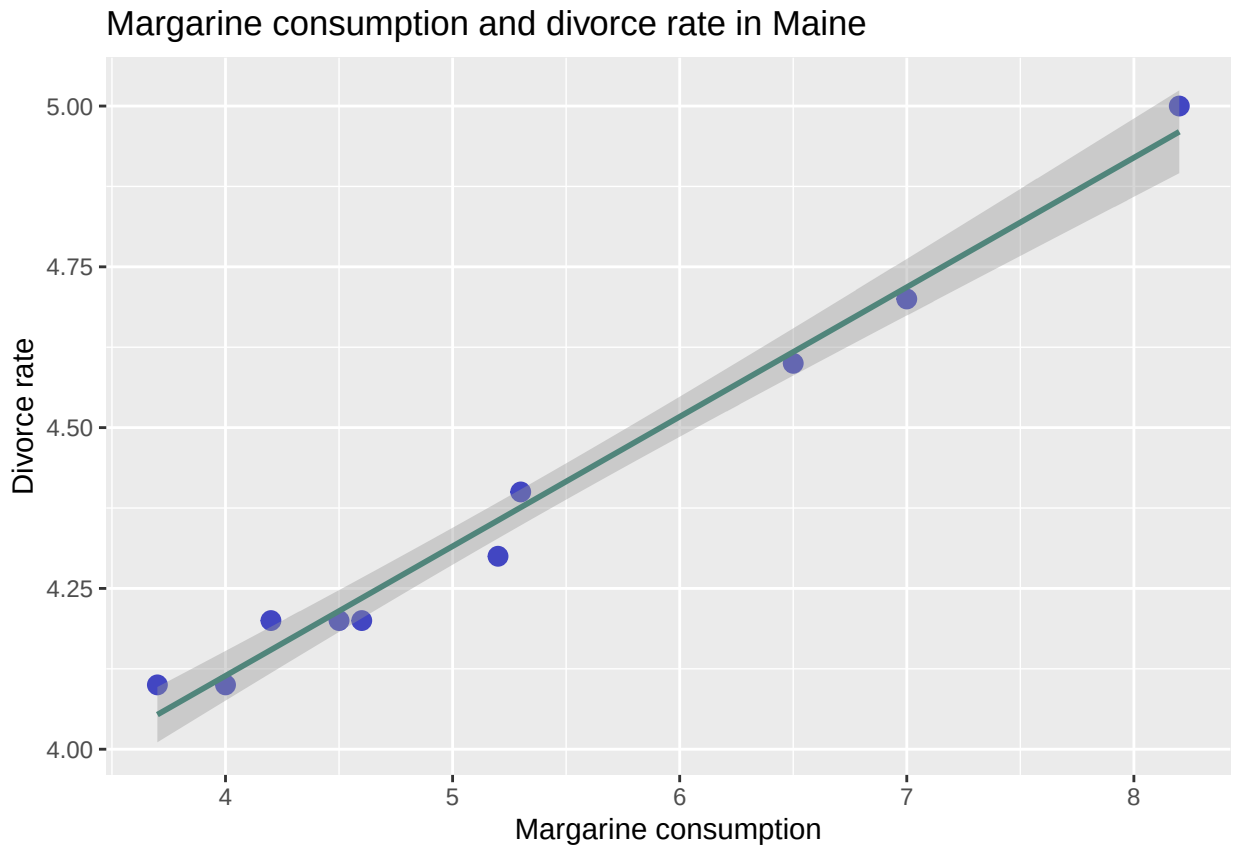
# Load relevant packages
library(dslabs)
library(dplyr)
library(ggplot2)
library(pacman)
```

```
# Check data
head(divorce_margarine)
```

```
##   divorce_rate_maine margarine_consumption_per_capita year
## 1                5.0                                8.2 2000
## 2                4.7                                7.0 2001
## 3                4.6                                6.5 2002
## 4                4.4                                5.3 2003
## 5                4.3                                5.2 2004
## 6                4.1                                4.0 2005
```

```
# Visualizing data

# Scatterplot with regressionline
ggplot(divorce_margarine, aes(x = margarine_consumption_per_capita, y = divorce_rate_maine)) +
  geom_point(size = 3, color = "#4246C2") +
  geom_smooth(method = "lm", color = "#50857B", se = TRUE) +
  labs(title = "Margarine consumption and divorce rate in Maine",
       x = "Margarine consumption",
       y = "Divorce rate")
```




```
# Testing for normality
shapiro.test(divorce_margarine$divorce_rate_maine)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  divorce_margarine$divorce_rate_maine
## W = 0.86135, p-value = 0.07916
```

```
shapiro.test(divorce_margarine$margarine_consumption_per_capita)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  divorce_margarine$margarine_consumption_per_capita
## W = 0.90531, p-value = 0.2503
```

```
#pearson correlation test for normality
cor.test(divorce_margarine$divorce_rate_maine,
         divorce_margarine$margarine_consumption_per_capita,
         method = "pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  divorce_margarine$divorce_rate_maine and divorce_margarine$margarine_consumption_per_capita
## t = 23.055, df = 8, p-value = 1.33e-08
## alternative hypothesis: true correlation is not equal to 0
```

```
## 95 percent confidence interval:
##  0.9676666 0.9983038
## sample estimates:
##       cor
## 0.9925585
```

The Shapiro-Wilk tests indicated that the data can be treated as approximately normal p = 0.08 & 0.25 and therefore p > 0.05. Given the approximate normality pearson correlation test for normality was used. Divorce rate was found to be significantly correlated with margarine consumption, r(8) = 0.99, p < .001, 95% CI [.97, 1.00]. Even though the correlation is significant it should not be interpreted as causal as other factors may explain the observed associations. Futhermore the sample set is very small (n=10) which limits the generalisability of the findings.

Part 2

Work with the 'GSSvocab' dataset from the 'car' package, focusing only on the year 1978 (subset that year and exclude missing values). Investigate how vocabulary test scores (vocab) are related to education (educ). Present the relationship in a plot, fit a regression model, and report results in APA format (coefficients, p-values, and R2), followed by a short practical interpretation.

Extend the analysis by making a new model, where you include whether a person is native- born (nativeBorn) as a predictor. Again, visualise and model the relationship. Consider whether education and native-born status interact in predicting vocabulary, and if so, fit an interaction model. For each model, report results in APA format and explain the practical meaning.

Finally, compare the models and discuss which one performs best based on p-values and R2, as well as the real-world interpretability of the results.

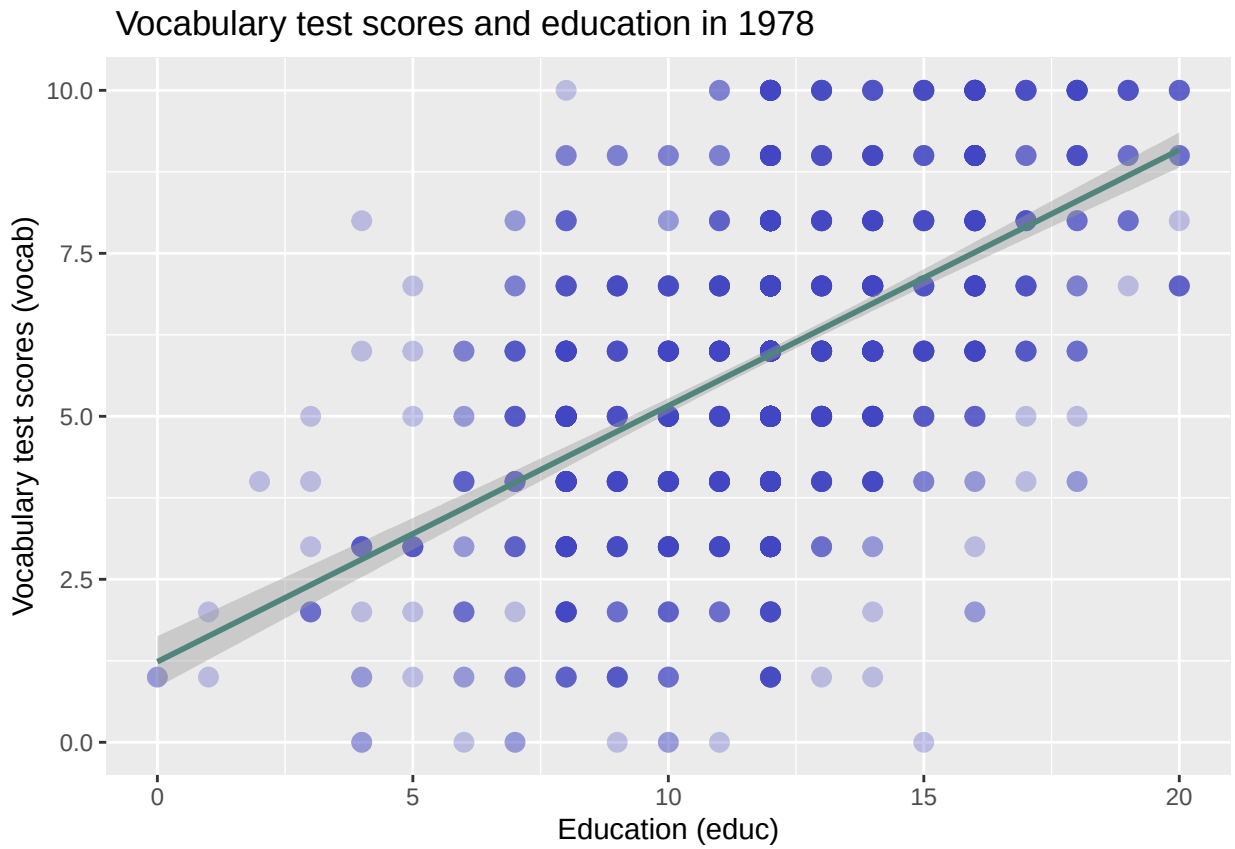
```r
# Loading Data
library(car)
df2 <- GSSvocab

# Focusing on the year 1978 and removing NA values
df2 <- filter(df2, year == 1978)
df2 <- na.omit(df2)



# Visualizing data

# Scatter plot with regression line
ggplot(df2, aes(x = educ, y = vocab)) +
  geom_point(size = 3, color = "#4246C2", alpha = 0.3) +
  geom_smooth(method = "lm", color = "#50857B", se = TRUE) +
  labs(title = " Vocabulary test scores and education in 1978",
       x = "Education (educ)",
       y = "Vocabulary test scores (vocab)")
```




```
# Making regression model
regression_model <-lm(vocab ~ educ, df2 )

summary(regression_model)
```

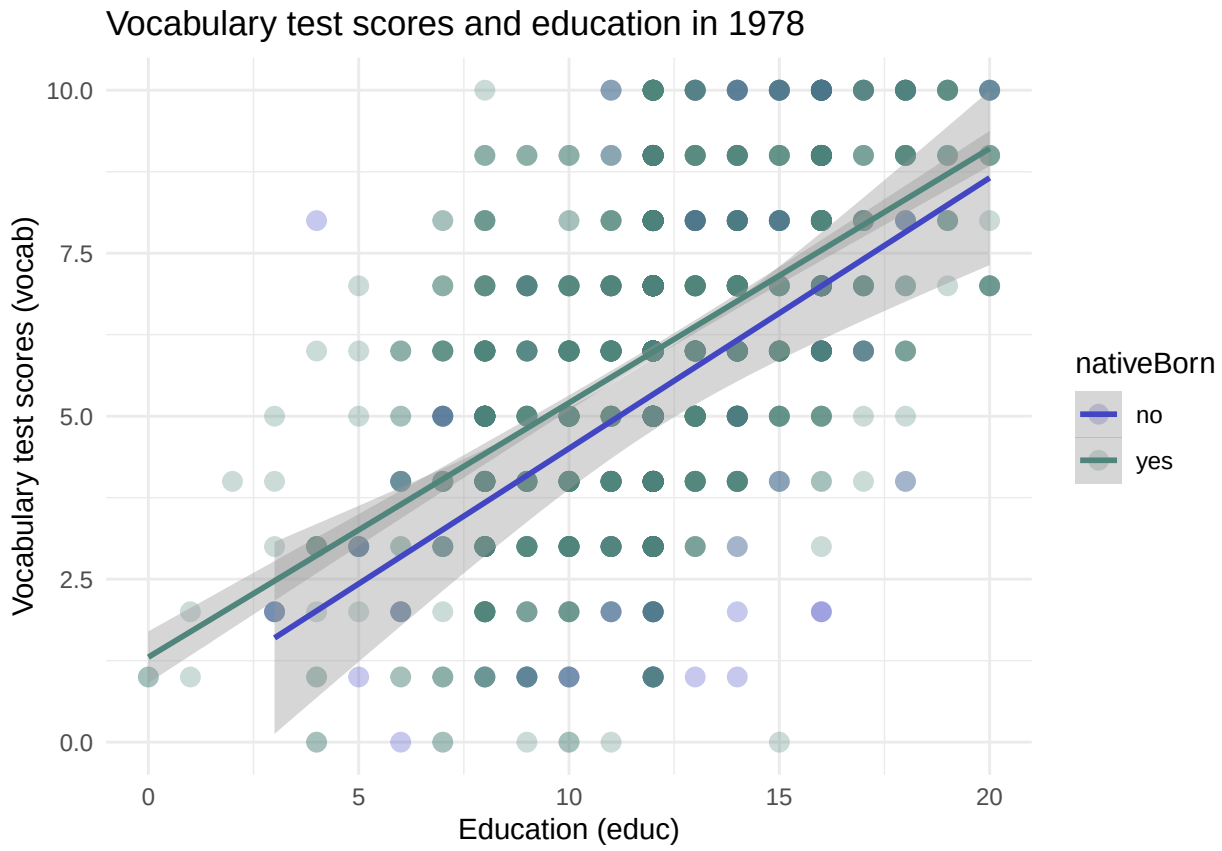
```
##
## Call:
## lm(formula = vocab ~ educ, data = df2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.1233 -1.1608  0.0542  1.0917  5.6243
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.23567    0.19957   6.192  7.7e-10 ***
## educ         0.39251    0.01606  24.443  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.885 on 1475 degrees of freedom
## Multiple R-squared:  0.2883, Adjusted R-squared:  0.2878
## F-statistic: 597.5 on 1 and 1475 DF,  p-value: < 2.2e-16
```

The linear regression model showed that education significantly predicted vocabulary test scores: b = 0.39, SE = 0.02, t(1475) = 24.44, p < .001, adjusted R-squared = 0.29. Looking at the the adjusted R-squared the model explains approximately 0.29% variance in vocabulary scores.

This indicates that for each additional year of education, vocabulary test scores increase with around 0.39 points on average.

```
# New regression model

# Scatterplot with regressionline
ggplot(df2, aes(x = educ, y = vocab, color = nativeBorn)) +
  geom_point(size = 3, alpha = 0.3) +
  geom_smooth(method = "lm") +
  scale_color_manual(values = c("#4246C2", "#50857B")) +
  labs(title ="Vocabulary test scores and education in 1978",
       x = "Education (educ)" ,
       y ="Vocabulary test scores (vocab)" ) +
  theme_minimal()
```




```
# New regression model including the variable native born
regression_model2 <- lm(vocab ~ educ + nativeBorn, df2 )
summary(regression_model2)
```

```
##
## Call:
## lm(formula = vocab ~ educ + nativeBorn, data = df2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.162 -1.200  0.015  1.231  5.803
##
```

```
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.62803    0.27651   2.271  0.02327 *
## educ           0.39222    0.01601  24.499  < 2e-16 ***
## nativeBornyes  0.65032    0.20551   3.164  0.00159 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.879 on 1474 degrees of freedom
## Multiple R-squared:  0.2931, Adjusted R-squared:  0.2921
## F-statistic: 305.6 on 2 and 1474 DF,  p-value: < 2.2e-16
```

The new linear regression model shows that education and "native born" both significantly predicted vocabulary test scores: education: b = 0.39, SE = 0.02, t(1475) = 24.49, p < .001, and native born: b = 0.65, SE = 0.21, t(1475) = 3.16, p < .001. The adjusted R-squared for the model = 0.29. This suggest that both education and being native-born are associated with higher vocabulary test scores.

```
regression_model3 <- lm(vocab ~ educ * nativeBorn, df2 )
summary(regression_model3)
```

```
##
## Call:
## lm(formula = vocab ~ educ * nativeBorn, data = df2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.1554 -1.2049  0.0149  1.2347  5.9857
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          0.35394    0.68780   0.515    0.607
## educ                 0.41510    0.05496   7.553 7.45e-14 ***
## nativeBornyes        0.95000    0.71855   1.322    0.186
## educ:nativeBornyes -0.02501    0.05745  -0.435    0.663
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.88 on 1473 degrees of freedom
## Multiple R-squared:  0.2932, Adjusted R-squared:  0.2917
## F-statistic: 203.7 on 3 and 1473 DF,  p-value: < 2.2e-16
```

The new linear regression model with an interactions did not improve the model fit and only educations is now significant p < .001. The adjusted R-squared for the model is also still around = 0.29.

The interaction between education and native-born status was not significant, b = -0.03, p = 0.66. While education remained a significant predictor (p < .001), nativeborn did not p = 0.19. The model explained approximately 29% of the variance (adjusted R^2 = 0.29).

This suggests that the relationship between education and vocabulary scores does not differ significantly between native-born and non-native-born individuals.

```
# Anova for comparing models
anova(regression_model, regression_model2, regression_model3)
```

```
## Analysis of Variance Table
##
## Model 1: vocab ~ educ
```

```
## Model 2: vocab ~ educ + nativeBorn
## Model 3: vocab ~ educ * nativeBorn
##   Res.Df    RSS Df Sum of Sq       F  Pr(>F)
## 1   1475 5241.8
## 2   1474 5206.5  1    35.371 10.0083 0.00159 **
## 3   1473 5205.8  1     0.670  0.1894 0.66344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

When comparing the models using Anovas, Model 2 appears to be the best-fitting model. This means that adding NativeBorn significantly approved the model, while the interaction did not.